

Microwave-Induced Grafting of Maleic Anhydride onto the Surface of a Polypropylene Film for High Adhesion to Polyamide 6

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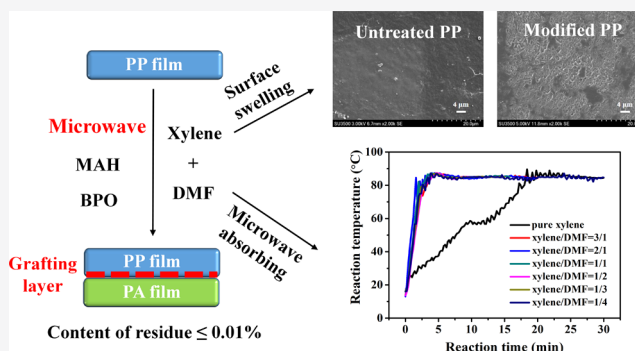
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ABSTRACT: A multilayer packaging technique in the food industry typically employs separate layers with unique mechanical and barrier properties to help preservation of food nutrition and flavors. Tie layers are generally required between adjacent functional layers due to their inherent poor adhesion, e.g., a maleic anhydride (MAH)-modified polypropylene (PP) tie layer is needed in between a PP-based moisture layer and a polyamide 6 (PA6)-based oxygen barrier layer, which not only complicates the manufacturing process but also poses potential health hazards from the migration of residues. This work puts forward a facile and effective approach by grafting MAH directly onto the surface of a PP film through microwave technology with xylene as a swelling agent and *N,N*-dimethylformamide as a microwave absorption agent. During the grafting process, microwave irradiation can shorten the reaction time by accelerating the decomposition of the initiator. Residual reactants can easily be removed, as they are only on the surface of the modified PP film, and the adhesive strength between the modified PP film and the PA6 film largely increased by 1 order of magnitude to 9.8 N/15 mm compared with that between the untreated PP film and the PA6 film.



1. INTRODUCTION

Packaging in the food industry is of utmost importance. Other than the primary function to store and protect the food material, it faces more stringent regulatory requirements in safety and sustainability as well as demands from consumers for better nutrition and flavor preservation and extended shelf life. Barely one material can suffice all of the requirements. A multilayer packaging technique combines the benefits of multiple materials in terms of their mechanical, moisture and oxygen barrier, and other functional properties through coextrusion and lamination.^{1,2} Polyethylene and polypropylene (PP) materials are widely used in the inner and outer layers in packaging to provide mechanical support, safe contact with food, and moisture guard.^{3,4} Polyamide (PA)⁵ and ethylene vinyl alcohol copolymer (EVOH)⁶ materials are used in the intermediate layer as oxygen barriers. A tie layer is typically needed between layers with materials of distinct polarities, e.g., PP and PA, to strengthen the adhesion and to maintain the mechanical integrity of the film.^{7,8}

Maleic anhydride (MAH)-grafted PP (PP-g-MAH) is frequently employed as the tie layer resin when PP has to be bonded to PA or EVOH.^{9,10} The anhydride functional group reacts with amine end groups in PA to form imides¹¹ and with alcohols in EVOH to form ester cross-links, while the main chain maintains good compatibility with the PP layer.¹² PP-g-MAH resin is industrially manufactured through free radical grafting by reactive extrusion.^{13,14} Impurities from residue

monomers and the degradation of PP may cause food contamination through chemical migration and permeation,¹⁵ film forming disturbance during the coextrusion process, and the neutralization of the terminal amine group of PA6, resulting in a severe decrease in adhesive strength.^{16–18}

Grafting of MAH onto the PP film through surface grafting methods offers an alternative route, which can potentially overcome the abovementioned problems encountered in a melt or solution process.¹⁹ First, active grafting sites, e.g., free radicals or ions, are mainly located on the surface of the film where anhydride functional groups graft onto;²⁰ with an adequate grafting extent of anhydride on the surface, a conventional tie layer may not be a necessity, which will greatly simplify the multilayer film manufacturing process and reduce the production cost. Second, residual monomers shall be easily removed, as the grafting reaction is restrained only on the surface of the film, unlike conventional processes in which residues are uniformly distributed in the bulk. In addition, the degradation of PP in the bulk can be inhibited since the

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possible β -scission reaction of PP chains only happens on the surface.

Commonly used surface grafting methods include solid-phase grafting,²¹ suspension grafting,²² and irradiation grafting.²³ Because of the low reaction temperature in solid-phase and suspension grafting processes, the reaction time is very long, which is not conducive from an industrial view.²⁴ Irradiation technologies, such as γ -ray, electron beams, plasma, etc., can produce a large number of active grafting sites onto polymer chains in a short time. However, the mechanical properties of the polymer are adversely affected to some degree under irradiation environments, and the cost of irradiation equipment and energy consumption are high.^{25,26} Compared with these, microwave irradiation is a special case, which can be effectively absorbed by polar substances but has no effect on nonpolar materials like PP.²⁷ Because the polar molecules of the initiator vibrate sharply with the high-frequency periodic change of the electric field under microwave irradiation, the decomposition rate of the initiator can be accelerated at low temperatures.^{28,29} As a result, the efficiency of the surface grafting reaction induced by microwave irradiation is high, while the bulk properties of PP are not influenced by surface modification.^{20,30}

Presently, several research studies used microwave-induced methods to graft MAH onto PP for the improvement of mechanical properties and wettability.^{31,32} In the grafting process, MAH dispersed inside the micropores of PP spherical powder or sheets can absorb microwave to generate high temperature and then cause itself and PP nearby to form free radicals; thus, MAH can graft onto the surface of PP. However, the approach requires a large amount of MAH since MAH is used as a microwave absorber. Moreover, PP can be degraded to a certain extent due to the instability of PP free radicals and rapid gasification of nearby MAH at high temperatures, which may affect its properties as an adhesive layer. To solve these problems, microwave-induced grafting should proceed at a mild and controllable temperature. In this work, MAH grafting onto the surface of PP films was proceeded under microwave irradiation using xylene as a swelling agent and *N,N*-dimethylformamide (DMF) as a microwave absorber in the presence of benzoyl peroxide (BPO) as an initiator. Through this method, both the grafting temperature of microwave-induced grafting and the dosage of MAH can dramatically decrease. The effects of different reaction conditions during the microwave grafting process of the PP film on its adhesive performance with a PA6 film were explored.

2. EXPERIMENTAL SECTION

2.1. Materials. Both the PP film and the PA6 film with a thickness of 0.2 mm were provided by Zhejiang Wazam New Materials Co., Ltd., China. Two commercial PP-g-MAH, CMG9801 and PO1020, were purchased from Fine-blend Polymer (Shanghai) Co., Ltd. and Dongguan City Yucheng Plastics Co., Ltd., respectively. MAH (AR), dichloromethane (CH_2Cl_2 , AR), chloroform (CHCl_3 , AR), methanol (AR), xylene (AR), DMF (AR), and acetone (AR) were purchased from Sinopharm Chemical Reagent Co., Ltd., China. BPO (containing 25% of water) was purchased from J&K Scientific Ltd., China. Prior to usage, the water in BPO should be removed by recrystallization (the detailed procedure can be seen in the SI).

2.2. Microwave-Induced Grafting of MAH on the PP Film. A measured amount of MAH was first dissolved in 200

mL of a mixed solvent of xylene and DMF, and then the solution was heated up to setting temperature. A PP film was soaked in the solution and then placed into a microwave reactor (Model FCMCR-3SX-T, Kerui Instrument Inc., China) to start the grafting reaction. A thermocouple through the inner wall of the reactor cavity can be put into the solution to measure the real-time temperature. The temperature was maintained at the setting value under microwave irradiation. Once the reaction was completed, the modified PP film was soaked in acetone for 12 h, washed, and then dried in a vacuum oven at 90 °C for 1 h to remove the residues.

2.3. Measurements. **2.3.1. Measurement of the Content of Residual Monomers in the Modified PP.** To quantify the content of residual MAH in the modified PP, a standard curve was first established to correlate the concentration of the MAH aqueous solution and ultraviolet (UV) absorbance at 230 nm (more details can be seen in the SI). Then, about 0.3 g of modified PP was swelled in a mixing solvent of 20 mL of water and 20 mL of CH_2Cl_2 in an autoclave at 125 °C for 90 min to extract the unreacted MAH into water. After the autoclave was cooled down to room temperature, 3 mL of the solution was taken from the aqueous phase of the upper layer for the measurement of UV absorbance at 230 nm using a UV–visible spectrophotometer (Model WFZ UV-3802, UNICO, China). Based on the standard curve mentioned above, the concentration of MAH in water could be obtained. So, the content of residual MAH (*W*) in the modified PP could be calculated by the following equation

$$W(\%) = \frac{A_s - b}{500 \times m_s \times a}$$

where A_s is the UV absorbance of extracted water at 230 nm, m_s (g) is the weight of the sample of the modified PP, and *a* and *b* are the coefficients in the linear fitting equation between the concentration of the MAH aqueous solution and UV absorbance.

2.3.2. Scanning Electron Microscope. A scanning electron microscope (SEM, model SU-3500, Hitachi, Japan) was used to characterize the surface morphology of the PP film at an acceleration voltage of 30 kV. Prior to observation, a layer of gold–palladium alloy was sprayed onto the surface of the film.

2.3.3. X-ray Photoelectron Spectrometer. An X-ray photoelectron spectrometer (XPS, model Escalab250Xi, Thermo Fisher Scientific Inc.) was used to analyze the elemental composition of the film surface with an energy analysis range of 0–5000 eV and an energy step of 6 meV. Al *K* α was used as the X-ray source. The binding energy was calibrated by a C 1s peak at 284.4 eV. After the test, the data of XPS spectra were normalized by the min–max normalization method.

2.3.4. Adhesive Strength. A modified PP film was cut into 12 cm $\times$ 5 cm samples and then laminated with a PA6 film of the same dimension in a thermal compressor (Model GT-7014-A50C, Gotech Testing Machine Inc., China) for 40 s at 220 °C and 4 MPa. Then, the composite film was cut into strips in a size of 100 mm $\times$ 15 mm to perform a T-peel test (Model Zwick/Roell Z020, Zwick, German) at a peeling rate of 20 mm/min. The maximum stress measured during the test is taken as the adhesive strength (unit: N/15 mm). At least three samples were tested each and the average values were calculated.

3. RESULTS AND DISCUSSION

3.1. Mechanism of Microwave-Induced Grafting of MAH onto the PP Film. For the grafting modification of PP, the common reaction mechanism is as follows: the initiators first decompose into small-molecule free radicals to attack the PP molecular chain and form PP macromolecular radicals, and then PP macromolecular radicals collide with the modified monomers to induce the grafting reaction. To increase the grafting degree of MAH onto the surface of the solid PP film, an effective approach is to increase the molecular chain mobility of the PP film surface by an organic solvent swelling slightly the surface layer but having no damage to the integrity of the PP film at the reaction temperature. In this work, xylene was selected as the swelling agent because it cannot dissolve PP but can slightly swell the surface of the PP film at the microwave reaction temperature. However, if only xylene is used as the reaction medium, it is ineffective in absorbing microwave due to its poor polarity, which leads to a very low heating rate under microwave, as shown in Figure 1. More

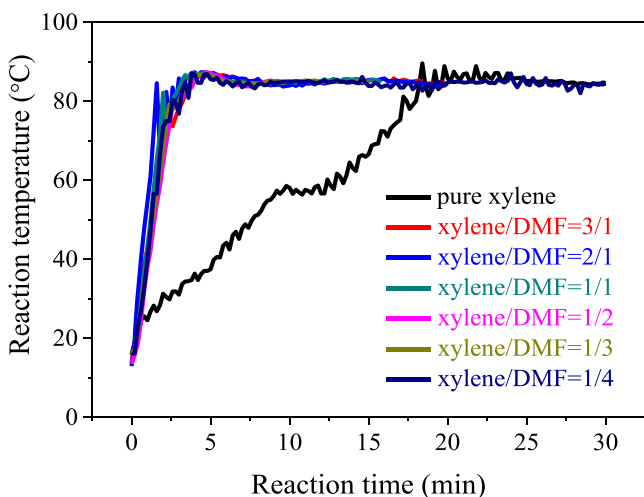


Figure 1. Temperature changes of the solutions with different ratios of xylene and DMF under microwave.

importantly, the temperature cannot be maintained stable. The addition of polar solvent DMF miscible with xylene dramatically improves the absorption efficiency to microwave, as the heating rate increases rapidly (Figure 1). Moreover, the temperature fluctuation is less than ± 1 °C after reaching the setting temperature.

Since the grafting reaction induced by microwave irradiation only occurs on the surface of the PP film, the grafting degree of MAH was too low to be analyzed quantitatively. Generally, the adhesive performance of the modified PP film with the PA6 film increases with the improvement of the grafting degree of MAH because it depends on the amount of the interfacial reaction between amine-terminated PA6 and the anhydride group on the PP film surface. To determine the optimal ratio of xylene and DMF, its effect on the adhesive strength between the modified PP film and the PA6 film is shown in Figure 2. When the ratio of xylene and DMF is 1/3, the adhesive strength is very low because the swelling effect of xylene on the surface of the PP film is not enough to promote the grafting reaction, which can be confirmed from Figure 3. Compared with the untreated PP film, the surface morphology of the PP film after microwave grafting in the xylene/DMF (1/3)

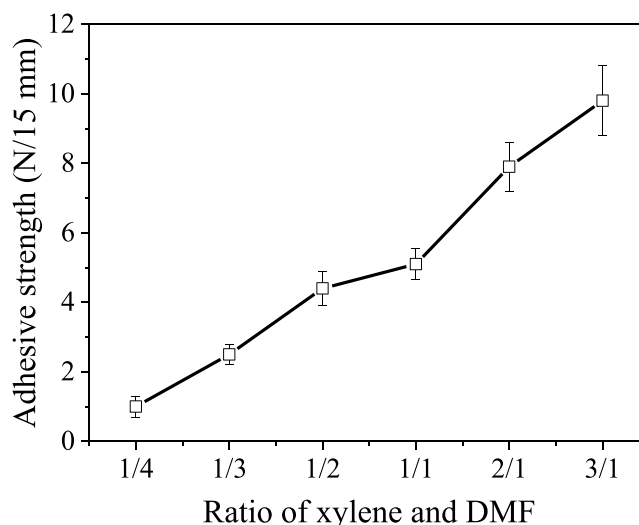


Figure 2. Effect of the ratio of xylene and DMF on the adhesive strength of the modified PP film with the PA6 film. The conditions for the modified PP film: temperature: 85 °C, reaction time: 30 min, and the concentrations of MAH and BPO: 0.03 and 0.003 g/mL, respectively.

solution changes very slightly. With the increase of the xylene proportion, some block structures stack on the surface and the roughness increases significantly, and the adhesive strength increases accordingly. Specifically, when the ratio of xylene and DMF is 3/1, the surface of the modified PP film is very rough with a swelling thickness of about 10 μm , as shown in Figure 3. Moreover, its adhesive strength with the PA6 film reaches 9.8 N/15 mm, as shown in Figure 2. From the XPS spectra shown in Figure 4, it can be further seen that compared with the untreated PP film and the modified PP film in the xylene/DMF (1/3) solution, the spectrum of the modified PP film in the xylene/DMF (3/1) solution shows an obvious new peak of O=C=O bond corresponding to MAH at 288.1 eV,³³ which also illustrates that the grafting degree of MAH on the PP film surface can be enhanced with the increase of the xylene proportion.

However, as the ratio of xylene and DMF further increases, the efficiency of microwave absorption reduces due to the increase in the content of xylene with low polarity, resulting in the fluctuation of the temperature of the reaction solution under microwave. Meanwhile, the swelling degree of the PP film also increases with the increase of xylene content, which easily causes the destruction of the integrity of the PP film. To avoid the above unfavorable phenomena, the ratio of xylene and DMF was selected at 2/1 for the following experiment.

To find out whether microwave irradiation plays other roles besides heating in the process of microwave-induced grafting, the adhesive properties of modified PP films prepared by microwave-induced grafting and thermal-induced grafting were compared. For thermal-induced grafting, the grafting reaction of the PP film is proceeded in the mixed solvent of xylene and DMF heated by an oil bath. Under the same grafting reaction conditions, the adhesive strength of the PA6 film and the modified PP film by thermal-induced grafting is only 0.8 N/15 mm, which is much lower than that by microwave-induced grafting (7.8 N/15 mm). This may imply that microwave irradiation can not only provide uniform heating energy but also promote the grafting reaction of PP. As a matter of fact, the half-life of BPO at 85 °C is more than 2 h, which is much

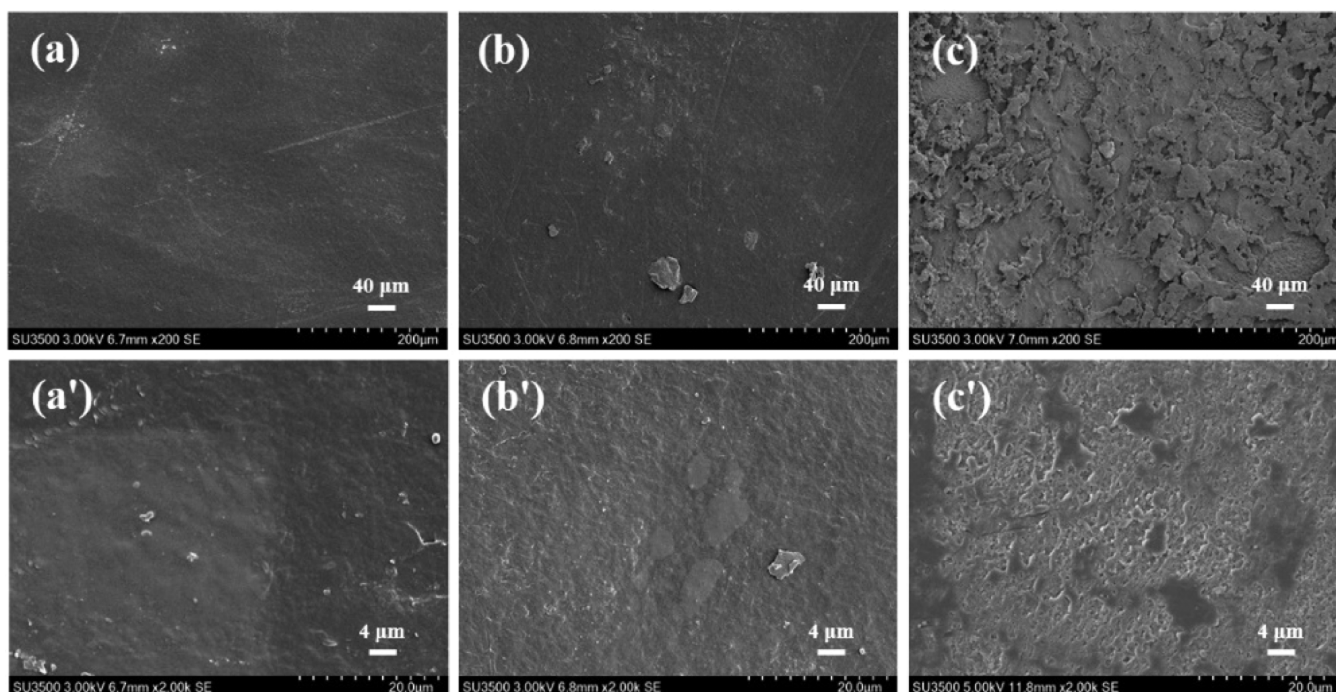


Figure 3. SEM images of the surface of PP films before (a, a') and after microwave in the xylene/DMF (1/3) solvent (b, b') and the xylene/DMF (3/1) solvent (c, c').

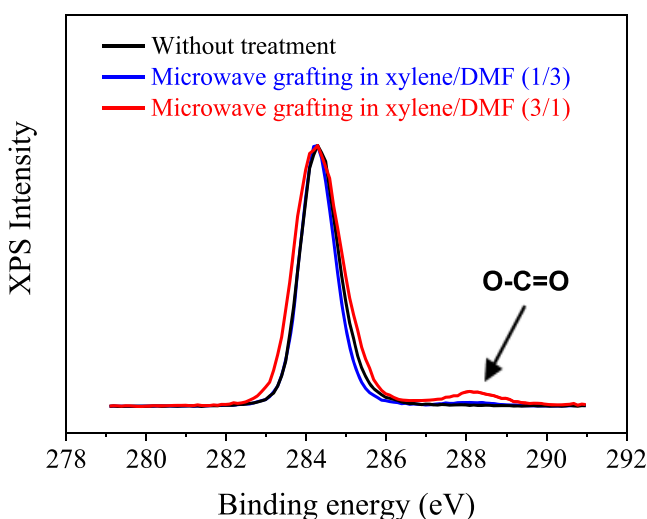


Figure 4. XPS spectra of the PP films before and after microwave grafting modification by MAH in the xylene/DMF (1/3) solvent and in the xylene/DMF (3/1) solvent.

longer than the reaction time (30 min). This may be the reason for the low grafting degree of MAH onto the PP film by thermal-induced grafting. In other words, microwave irradiation can accelerate the decomposition of initiators to increase the grafting degree of MAH onto the PP film. A similar conclusion was also obtained by Zhao et al.³⁴ that at room temperature (21 °C), more than 9% of BPO decomposed in 30 min under electromagnetic irradiation of 130 MHz, while only less than 0.2% of BPO decomposed without irradiation.

In general, with the help of toluene as the swelling agent and DMF as the microwave absorber, both the chain mobility on the PP film surface and the decomposition rate of the initiator are enhanced by microwave-induced grafting. Thus, the

grafting reaction of MAH onto the PP film is promoted and the required reaction time is greatly shortened.

3.2. Analysis of Residual Unreacted MAH in the Modified PP Film. The content of residual unreacted MAH in the modified PP film by the microwave-induced grafting method is shown in Table 1. The contents of residual MAH in

Table 1. Contents of Residual MAH in PP-g-MAH before and after Soaking in Acetone

sample	preparation method	contents of residual MAH [%]	
		before soaking	after soaking
self-made	microwave-induced grafting	0.036 ± 0.004	0.012 ± 0.002
CMG9801	melt grafting	0.18 ± 0.02	0.13 ± 0.04
PO1020	melt grafting	0.083 ± 0.010	0.083 ± 0.005

two kinds of commercial PP-g-MAH produced by reactive extrusion are presented for comparison. From Table 1, it can be clearly seen that the content of residual MAH in the former is lower than those in the latter two. Moreover, after being soaked in acetone at room temperature for 12 h, the content of residual MAH for the former can drop significantly from 0.04 to 0.01%. However, the residual contents for the latter two are still relatively high, about 0.18 and 0.08%, respectively. Therefore, compared with the conventional melt grafting method, the microwave-induced grafting approach can not only reduce the content of residual monomers but also remove the residual monomers more easily because the unreacted MAH is only on the surface layer of the modified PP film.

To explore whether the residual MAH can hinder the adhesion between the modified PP film and the PA6 film, the modified PP films were first covered with MAH by immersing them in MAH solutions with different concentrations. The FTIR spectra of these treated PP films are shown in Figure 5. The signal intensities of the carbonyl group of MAH at 1780

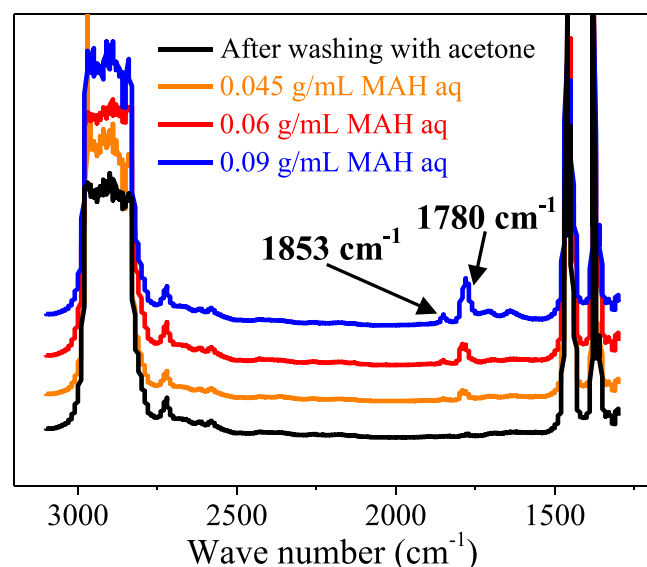


Figure 5. FTIR spectra of modified PP films after soaking in MAH solutions.

and 1853 cm^{-1} are enhanced with the increase of the concentration of the MAH solution. To be specific, after being soaked in the MAH solution with concentrations of 0.045, 0.06, and 0.09 g/mL, the contents of residual MAH in the modified PP films increase to 0.04, 0.06, and 0.09%, respectively. As shown in Table 2, the adhesive strength

Table 2. Contents of Residual MAH and its Adhesive Strengths with the PA6 Film of Modified PP Films before and after Soaking in MAH Solutions with Different Concentrations

concentrations of the MAH solution [g/mL]	before soaking in acetone		after soaking in acetone	
	content of residual MAH [%]	adhesive strength [N/15 mm]	content of residual MAH [%]	adhesive strength [N/15 mm]
0	0.01	9.8	0.01	9.5
0.045	0.04	9.4	0.01	9.9
0.06	0.06	7.4	0.01	10.4
0.09	0.09	6.3	0.01	10.0

between the modified PP film and the PA6 film gradually decreases with the increase of the content of residual MAH. After being soaked in acetone, the contents of residual MAH decrease to 0.01%, and their adhesive strengths with the PA6 film return to the original level. Obviously, the presence of the residual monomer in modified PP has a negative effect on its adhesive performance. Therefore, the microwave-induced grafting of MAH onto the PP film is beneficial to reduce the content of residual MAH and improve the adhesive strength.

3.3. Influences of Reaction Conditions on the Adhesive Strength of the Modified PP Film. **3.3.1. Reaction Temperature.** The effect of microwave reaction temperature on the adhesive strength of the modified PP film with the PA6 film is shown in Figure 6. When the reaction temperature is low, the concentration of free radicals is low due to slow decomposition of BPO. Moreover, the swelling effect of xylene is not obvious. As a result, the grafting reaction is difficult to proceed, which leads to weak adhesive strength. As the reaction temperature increases, the decomposition of the initiator accelerates, and the swelling degree of the PP film

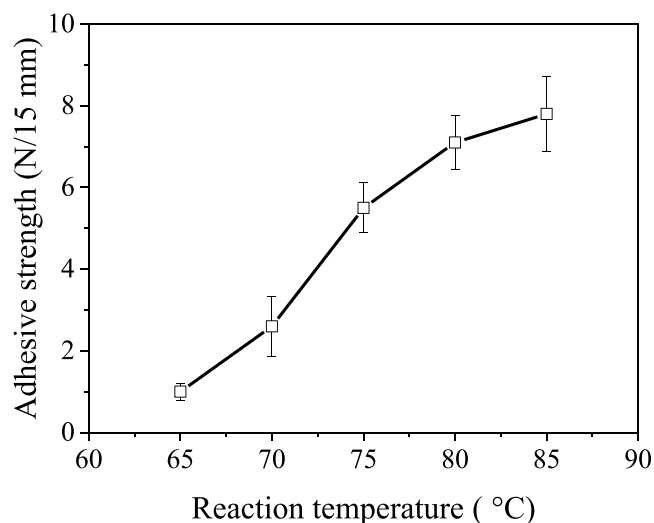


Figure 6. Effect of reaction temperature on the adhesive strength of the modified PP film with the PA6 film. The ratio of xylene and DMF: 2/1, reaction time: 30 min, and the concentrations of MAH and BPO: 0.03 and 0.003 g/mL, respectively.

surface by xylene is also improved, which is beneficial to increase the grafting degree of MAH onto the PP film for enhancing its adhesive strength with the PA6 film. However, when the reaction temperature increases to above $90\text{ }^{\circ}\text{C}$, the PP film will be destroyed due to the dissolution in xylene. Therefore, the optimum microwave reaction temperature is $85\text{ }^{\circ}\text{C}$.

3.3.2. Reaction Time. Figure 7 presents the influence of microwave reaction time on the adhesive strength of the

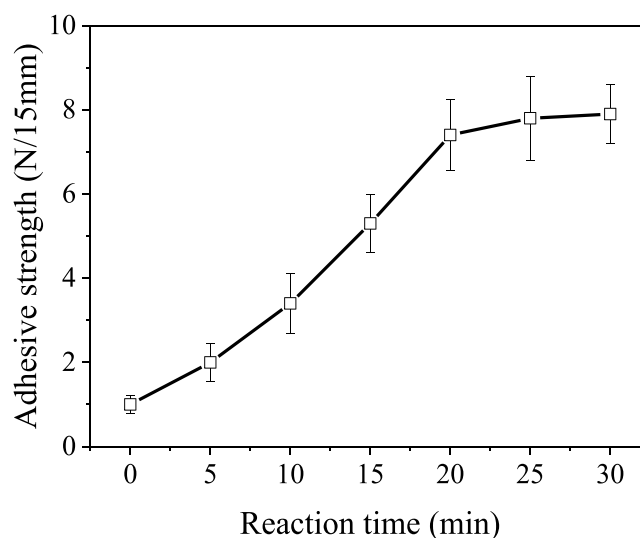


Figure 7. Effect of reaction time on the adhesive strength of the PP film with the PA6 film. The ratio of xylene and DMF: 2/1, reaction temperature: $85\text{ }^{\circ}\text{C}$, and the concentrations of MAH and BPO: 0.03 and 0.003 g/mL, respectively.

modified PP film with the PA6 film. From Figure 7, it can be seen that the adhesive strength first increases with the reaction time and then gradually reaches a plateau. At the beginning of the reaction, the concentration of radicals constantly increases to improve the grafting degree of MAH, which leads to the increase of the adhesive strength of the modified PP film with

the PA6 film accordingly. After 20 min of reaction, due to the consumption of the monomer and the initiator, the probability of effective collision between different molecules reduces and the grafting reaction gradually stops, and the adhesive strength of the modified PP film also reaches the maximum value.

3.3.3. MAH Concentration. The effect of MAH concentration in the mixed solvent of xylene and DMF on the adhesive strength of the modified PP film with the PA6 film is shown in Figure 8. As the MAH concentration increases, more

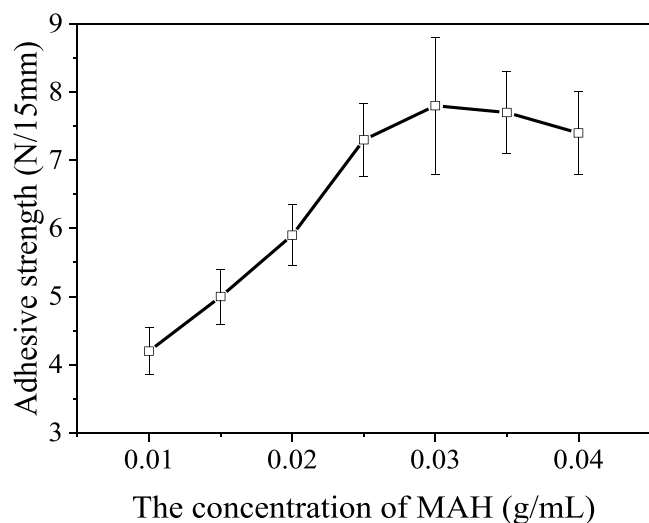


Figure 8. Effect of MAH concentration on the adhesive strength of the PP film with the PA6 film. The ratio of xylene and DMF: 2/1, reaction temperature: 85 °C, reaction time: 30 min, and the concentration of BPO: 0.003 g/mL.

MAH can be grafted onto the surface of the PP film to improve its adhesive strength with the PA6 film. For example, when the MAH concentration increases from 0.01 to 0.03 g/mL, the adhesive strength enhances from 4.2 to 7.8 N/15 mm. When the MAH concentration continuously increases, the density of monomer radicals increases and homopolymerization between MAH also increases, which consumes effective radicals in solution. The consumption of free radicals by excessive MAH was also observed by Zhang et al.³⁵ in grafting MAH onto PP. As a result, the grafting reaction is hindered. Therefore, when the MAH concentration in solution is too high, the adhesive strength of the PP film decreases slightly.

3.3.4. Initiator Concentration. Figure 9 shows the influence of initiator concentration on the adhesive strength of the modified PP film with the PA6 film. The adhesive strength gradually increases with the increase of initiator concentration and reaches the maximum value of 8 N/15 mm when the initiator concentration is 0.004 g/mL. As the concentration of BPO increases, more PP radicals are generated, which is beneficial to the proceeding of the grafting reaction. However, when the initiator concentration further increases, the probability of the coupling termination reaction also increases, which hinders the MAH grafting reaction on the surface of the PP film. A similar conclusion was also obtained by Galia et al.³⁶ in grafting of MAH onto PP assisted with supercritical carbon dioxide. As a result, the adhesive strength of the modified PP film with PA6 decreases.

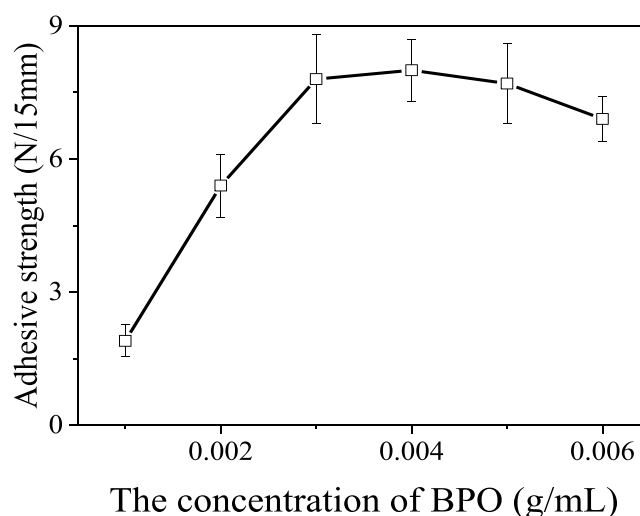


Figure 9. Effect of BPO concentration on the adhesive strength of the PP film with the PA6 film. The reaction conditions for the modified PP film: the ratio of xylene and DMF: 2/1, reaction temperature: 85 °C, reaction time: 30 min, and the concentration of MAH: 0.03 g/mL.

4. CONCLUSIONS

In this work, using xylene as the swelling agent and DMF as the microwave absorber, MAH was successfully grafted onto the surface of the PP film under microwave irradiation. The decomposition of peroxide can be accelerated by microwave irradiation to promote the grafting reaction of MAH onto the PP film. After the grafting reaction, the content of residual MAH in the modified PP film is much lower than that of PP-g-MAH obtained by reactive extrusion. Due to the introduction of MAH onto the surface of the PP film, its adhesive strength with the PA6 film is enhanced significantly from 1.0 N/15 mm to a maximum value of 9.8 N/15 mm. By investigating the influence of different factors on the adhesive strength of the modified PP film with the PA6 film, the optimum reaction conditions of microwave-induced grafting of MAH onto the surface of the PP film are determined as follows: the ratio of xylene and DMF is 2/1, the reaction temperature is 85 °C, the reaction time is 30 min, and the concentrations of BPO and MAH are 0.003 and 0.03 g/mL, respectively.

However, from an industrial view, the usage of a large amount of organic solvent limits the application of this method because of the difficulties in discharge or recovery of waste solution. Instead of that, water is also an effective microwave absorber with no harmful effects on the environment; the microwave grafting reaction in aqueous solution will be investigated in our next work. Besides, the technology development for accurate characterization of the microscopic properties of modified surfaces, such as grafting density or grafting chain length, will help researchers to deeply understand the mechanism of adhesion between polymers and broaden the application of the microwave-induced grafting method in more reaction systems.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.iecr.1c03701>.

Procedures of refinement of BPO by recrystallization; establishment of the standard curve for measuring the content of MAH in the PP film; concentration and UV absorbance of standard MAH solutions (Table S1); and standard curve of the UV absorbance at 230 nm and the concentration of MAH (Figure S1) (PDF)

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Notes

The authors declare no competing financial interest.

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